

AI & Prompt Engineering

Assignment_1

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Track: Web & UI

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✓ **Selected Use Case**

Software Developer Assistant

🎯 **Problem Statement**

Junior developers often struggle to design scalable, secure authentication systems in React applications, especially when integrating frontend and backend securely. The goal is to use Generative AI as a Software Developer Assistant to guide, reason, refine, and improve a solution through structured prompting.

◇ **Prompt 1 – RTF + RISEN Techniques**

used:

- ✓ RTF (Role-Task-Format)
- ✓ RISEN (Role, Instructions, End goal, Narrowing)

You are a Senior Software Engineer and System Architect acting as a Software Developer Assistant.

Role:

Guide developers in designing secure, scalable frontend architectures.

Instructions:

I am building a React application that requires user authentication using JWT.

I want to understand the best architectural approach for handling authentication securely on the frontend.

End Goal:

Provide a high-level authentication architecture suitable for a production-grade React application.

Narrowing:

Focus only on frontend concerns (state management, token handling, security best practices).

Do not include backend implementation details yet.

What this demonstrates:

- Clear role definition
- Clear scope
- No ambiguity
- Professional instruction clarity

◆ **Prompt 2 — Iterative Improvement**

Iterate on the previous architecture under these new constraints:

- The application must support multi-tab synchronization
- Authentication state must survive page reloads without storing access tokens
- The solution should remain XSS-resilient

Modify only what is necessary.

Clearly state what changed and why.

What this proves

- ✓ Iteration based on prior output
 - ✓ Constraint-based refinement
 - ✓ Architectural thinking
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◇ **Prompt 3 — Final Synthesis (End Goal Closure)**

Provide a final summary of the frontend authentication architecture in:

- 5 concise bullet points
- One real-world analogy
- One checklist a developer can follow before deployment

This is for

- Real-world example to relate that to real projects
- Checklist that can be followed to ensure secure and organized implementation